

cancer therapy¹ should be balanced with their potential to persist despite potent therapy.

Rachel M. Reardon, BS,^{a,b} Katherine L. Perlman, MPH,^{a,c} Maria S. Asdourian, MPhil,^{a,b} Nishi Shah, BS,^a Ted V. Jacoby, BS,^a Leah L. Thompson, MD,^a Kerry L. Reynolds, MD,^d Yevgeniy R. Semenov, MD, MA,^a and Steven T. Chen, MD, MPH, MS-HPed^{a,b}

From the Department of Dermatology, Massachusetts General Hospital, Boston, Massachusetts^a; Harvard Medical School, Boston, Massachusetts^b; University of Illinois College of Medicine, Chicago, Illinois^c; and Division of Hematology and Oncology, Department of Medicine, Massachusetts General Hospital Cancer Center, Boston, Massachusetts.^d

Authors Reardon and Perlman contributed equally to this work.

Funding sources: Dermatology Foundation Medical Dermatology Career Development Award (Dr Chen).

Elements of this work to be included in a presentation at the World Congress of Dermatology, Singapore, July 7 2023.

Patient consent: Not applicable.

IRB approval status: Approved by the Institutional Review Board of Mass General Brigham (#2017P000501).

Key words: dermatologic immune-related adverse events; immune checkpoint inhibitor; immunotherapy; oncology.

Correspondence to: Steven T. Chen, MD, MPH, MS-HPed, Department of Dermatology, Massachusetts General Hospital and Harvard Medical School, 50 Staniford St, Ste 200, Boston, MA 02114

E-mail: stchen@partners.org

Conflicts of interest

Dr Chen reports receiving personal fees from Argenx, Pfizer, Novartis, ScholarRock, and VisualDx outside of the submitted work. Author Reardon, Author Perlman, Author Asdourian, Author Shah, Author Jacoby, Dr Thompson, Dr Reynolds, and Dr Semenov have no conflicts of interest to declare.

REFERENCES

1. Thompson LL, Chang MS, Polyakov NJ, et al. Prognostic significance of cutaneous immune-related adverse events in patients with melanoma and other cancers on immune checkpoint inhibitors. *J Am Acad Dermatol*. 2022;86(4):886-889. <https://doi.org/10.1016/j.jaad.2021.03.024>

2. Sibaud V. Dermatologic reactions to immune checkpoint inhibitors: skin toxicities and immunotherapy. *Am J Clin Dermatol*. 2018;19(3):345-361. <https://doi.org/10.1007/s40257-017-0336-3>
3. Asdourian MS, Shah N, Jacoby TV, et al. Evaluating patterns of co-occurrence between cutaneous and noncutaneous immune-related adverse events after immune checkpoint inhibitor therapy. *J Am Acad Dermatol*. 2023;88(1):246-249. <https://doi.org/10.1016/j.jaad.2022.05.015>
4. U.S. Department of Health and Human Services. Common terminology criteria for adverse events (CTCAE) version 5.0. Accessed February 23, 2022. https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/CTCAE_v5_Quick_Reference_8.5x11.pdf
5. Sadik CD, Langan EA, Gutzmer R, et al. Retrospective analysis of checkpoint inhibitor therapy-associated cases of bullous pemphigoid from six German dermatology centers. *Front Immunol*. 2021;11:588582. <https://doi.org/10.3389/fimmu.2020.588582>

<https://doi.org/10.1016/j.jaad.2024.01.070>

Executive summary: Consensus treatment guidelines for the use of methotrexate for inflammatory skin disease in pediatric patients



To the Editor: Methotrexate (MTX) is an inexpensive and readily available medication that is used off-label for many inflammatory skin conditions in pediatric patients, with substantial variation in dosing and monitoring. To support optimal management of pediatric inflammatory skin diseases with low-dose MTX, a 27-member expert committee convened to establish evidence-based consensus-driven guidelines.¹ The following 5 major subjects were assessed: (1) indications and contraindications, (2) dosing, (3) interactions with immunizations and medications, (4) adverse effects, and (5) monitoring. An executive summary is presented below. Open access to the full-length guidelines is available; see Seigfried et al.¹

The committee included pediatric and general dermatologists, pediatric rheumatologists, and participants from industry and patient advocacy groups. An extensive literature search was initiated in July 2019 using keywords from clinical questions developed for each of the 5 subjects. Preliminary consensus statements were drafted based on the available literature and expert opinions from the committee. An online survey platform and modified Delphi process were conducted to achieve consensus, predefined as ≥70% of participants rating a statement as 7 to 9 on a 0 to 9 Likert scale. The Strength of Recommendation Taxonomy system² was used to rate the strength of recommendations and quality of available data.

Table 1. Key consensus statements to guide MTX use for inflammatory skin disease in children

Recommendation	Respondent agreement, %	SORT ^a
Dosing		
While both weight-based dosing and body surface area-based dosing have been used for MTX in pediatric patients, weight-based dosing may be easier to calculate and has been safely used in many studies.	88	AI
Maximum dose of MTX in published studies varies widely, but for inflammatory skin disease the maximum dose is considered 1 mg/kg, not exceeding 25 mg/wk.	92	CIII
Parenteral administration of MTX likely results in higher bioavailability and may increase efficacy.	88	CIII
Test doses are not necessary for pediatric patients starting low-dose MTX for inflammatory skin disease.	85	CIII
Onset of MTX effect is often 8-12 wk for atopic dermatitis, psoriasis, and lichen planus but may be closer to 12-16 wk for alopecia areata and morphea. Time to maximal effect may be longer.	92	CIII
MTX can be discontinued abruptly without adverse effects, other than risk of disease worsening.	88	CIII
The rate of taper should be based on disease severity, risk of disease worsening, and/or patient and parent/caregiver preference.	80	CIII
Interactions		
Concomitant use of MTX and trimethoprim/sulfamethoxazole is not contraindicated.	75	CIII
NSAIDs or salicylates can be given concurrently with low-dose MTX in children with normal renal function.	84	BII
Adverse effects		
Drug-related adverse effects of MTX including gastrointestinal effects, oral mucositis, fatigue, and headache are typically mitigated with folic acid supplementation.	92	BII
The use of folic acid is recommended during treatment with MTX, and daily dosing of 1 mg/d is recommended over once-weekly dosing for most patients.	92	CIII
The following information is pertinent in any patient on MTX with elevated liver enzymes: recent symptoms of viral illness, MTX dosing within 24 h prior to phlebotomy, recent administration of other medications (eg, acetaminophen), recent alcohol consumption.	96	CIII
MTX should be temporarily discontinued when liver enzymes lab results are $\geq 3 \times$ ULN for 2 consecutive mo.	92	CIII
Discontinue or decrease the dose of MTX if total white blood cell count is $<3 \times 10^9/L$, absolute neutrophil count is $<10 \times 10^9/L$, or platelet count is $<100,000/L$.	92	CIII
Counseling and mitigation strategies regarding the risks of fetal loss and teratogenicity should be discussed and implemented for individuals of childbearing potential taking MTX.	100	BII

MTX, Methotrexate; NSAID, nonsteroidal antiinflammatory drug; SORT, Strength of Recommendation Taxonomy; ULN, upper limits of normal.

^aSORT rating was used to rate the strength of each recommendation and quality of available data upon which each recommendation was rated as follows: A. Recommendation based on consistent and good quality patient-oriented evidence. B. Recommendation based on inconsistent or limited quality patient-oriented evidence. C. Recommendation based on consensus, opinion, case studies, or disease-oriented evidence. I. Good-quality patient-oriented evidence (ie, measures outcomes that matter to patients: morbidity, mortality, symptom improvement, cost reduction, and quality of life). II. Limited-quality patient-oriented evidence. III. Other evidence, including consensus guidelines, usual practice, opinion, disease-oriented evidence (ie, measures intermediate, physiologic, or surrogate endpoints that may or may not reflect improvements in patient outcomes), and case series for studies of diagnosis, treatment, prevention, or screening.²

Forty-six recommendations reached consensus, including the following:

- Lack of Food and Drug Administration approval does not preclude the clinical decision to prescribe MTX to pediatric patients with inflammatory skin disease.
- MTX is useful for treating pediatric patients with morphea, psoriasis, dermatomyositis, atopic dermatitis, lupus, sarcoidosis, and alopecia areata, and for prevention of neutralizing

antibodies associated with biologic therapy. The committee recognized other conditions that can be effectively treated with MTX supported by limited clinical data and personal experience.

- While dosing based on either weight or body surface area can be used, weight-based dosing was recommended for ease of calculation in clinical practice.
- There was agreement on a maximum maintenance dose of 1 mg/kg, up to 25 mg/wk;

Table II. Recommendations to guide MTX monitoring for inflammatory skin disease in children

Recommendation	Respondent agreement, %	SORT ^a
Baseline laboratory monitoring for all patients using MTX should include: • Complete blood count • White blood cell differential • Alanine aminotransferase • Aspartate aminotransferase • Serum creatinine	92-100	CIII
Ongoing laboratory monitoring should be done after 1st mo and then every 3-4 mo: • Complete blood count • White blood cell differential • Alanine aminotransferase • Aspartate aminotransferase • Serum creatinine	75—88	CIII
Baseline pregnancy testing should be performed for all patients of childbearing potential who are sexually active and not using contraception.	88	BII
Tuberculosis testing is only recommended for patients at risk for exposure.	83	CIII
Labs should be repeated sooner than usual for liver enzyme elevations 2-3 × ULN.	100	CIII
Liver biopsy is not indicated for routine monitoring of pediatric patients taking low-dose MTX.	96	BII
Pulmonary function tests are only indicated for patients with pulmonary symptoms.	100	CIII

MTX, Methotrexate; SORT, Strength of Recommendation Taxonomy; ULN, upper limits of normal.

*The Strength of Recommendation Taxonomy system² was used to rate the strength of each recommendation as: A) recommendation based on consistent and good-quality patient-oriented evidence; B) recommendation based on inconsistent or limited-quality patient-oriented evidence; or C) recommendation based on consensus, opinion, case studies, or disease-oriented evidence. The quality of available data upon which each recommendation was based was rated as: I) good-quality patient-oriented evidence (ie, measures outcomes that matter to patients: morbidity, mortality, symptom improvement, cost reduction, and quality of life); II) limited-quality patient-oriented evidence; or III) other evidence, including consensus guidelines, usual practice, opinion, disease-oriented evidence (ie, measures intermediate, physiologic, or surrogate endpoints that may or may not reflect improvements in patient outcomes), and case series for studies of diagnosis, treatment, prevention, or screening.

however, consensus was not achieved about an optimal starting dose, corresponding to the wide dosing range reported in the literature.

- MTX can be administered orally or subcutaneously, but the parenteral route likely enhances bioavailability and may increase efficacy.
- An initial “test dose” (<7.5 mg) of MTX is not needed for pediatric patients with inflammatory skin disease.
- Consider the impact of MTX on immunization and coadministered medications; multiple specific recommendations were made in this category.
- Although MTX is generally well tolerated, it is important to counsel patients and caregivers about potential adverse effects, most commonly gastrointestinal. Daily supplementation with 1 mg/d oral folic acid can minimize gastrointestinal discomfort.

Table I includes a selection of key consensus statements addressing dosing, interactions and adverse effects. Table II lists specific recommendations for monitoring patients on MTX.

Even in the era of targeted therapies, MTX continues to play an important role in the management of inflammatory skin diseases. These consensus-based

recommendations are intended to inform dosing, monitoring, and mitigation of potential side effects of MTX for treatment of pediatric inflammatory skin disease.

The authors would like to thank Jennifer C Jaworski, MS, BCMAS, CMPP, Umar Sheikh, MD, and Rita Sieracki, MLS for their contributions to this work. The authors would also like to thank the Pediatric Dermatology Research Alliance (PeDRA) Best Practices Task Force for their review of the manuscript.

Heather A. Brandling-Bennett, MD,^{a,b} Lisa M. Arkin, MD,^c Yvonne E. Chiu, MD,^d Adelaide A. Hebert, MD,^e Jeffrey P. Callen, MD,^f Leslie Castelo-Soccio, MD,^g Dominic O. Co, MD,^c Kelly M. Cordoro, MD,^b Megan L. Curran, MD,ⁱ Austin M. Dalrymple, MD,^{j,k} Carsten Flohr, MD,^l Ken B. Gordon, MD,^d Diane Hanna, DNP,^m Alan D. Irvine, MD,ⁿ Susan Kim, MD,^b A. Yasmine Kirkorian, MD,^{o,p} Irene Lara-Corrales, MD,^q Jill Lindstrom, MD,^r Amy S. Paller, MD,^s Melissa Reyes, MD,^t Wendy Smith Begolka, MBS,^u Wynniss L. Tom, MD,^{v,w} Abby S. Van Voorbees, MD,^x Ruth Ann Vleugels, MD,^{y,z} Lara Wine Lee, MD,^{aa} Olivia Davies, MD,^d and Elaine C. Siegfried, MD,^{i,k}

From the Seattle Children's Hospital, Seattle, Washington^a; Division of Dermatology, Department of Pediatrics, University of Washington, Seattle, Washington^b; Department of Dermatology, University of Wisconsin School of Medicine and Public Health, Madison, Wisconsin^c; Department of Dermatology, Medical College of Wisconsin, Milwaukee, Wisconsin^d; Department of Dermatology, UTHealth McGovern Medical School at Houston, Houston, Texas^e; Department of Dermatology, University of Louisville School of Medicine, Louisville, Kentucky^f; Dermatology Branch, National Institutes of Arthritis and Musculoskeletal and Skin Diseases, National Institutes of Health, Bethesda, Maryland^g; Department of Dermatology, University of California, San Francisco, San Francisco, California^h; Division of Rheumatology, Department of Pediatrics, Children's Hospital Colorado, University of Colorado School of Medicine, Aurora, Coloradoⁱ; Division of Rheumatology, Department of Pediatrics, Saint Louis University School of Medicine, St. Louis, Missouri^j; SSM Health Cardinal Glennon Children's Hospital, St. Louis, Missouri^k; Department of Paediatric Dermatology, St John's Institute of Dermatology, Guy's & St Thomas' NHS Foundation Trust, London, UK^l; Modern Dermatology, Shawnee, Kansas^m; Department of Paediatric Dermatology, Trinity College Dublin, Dublin, Irelandⁿ; Department of Dermatology, George Washington University School of Medicine & Health Sciences, Washington, District of Columbia^o; Children's National Hospital, Washington, District of Columbia^p; Department of Pediatrics, Hospital for Sick Children, University of Toronto, Toronto, Ontario, Canada^q; Walter Reed National Military Medical Center, Bethesda, Maryland^r; Department of Dermatology, Northwestern University Feinberg School of Medicine, Chicago, Illinois^s; Food and Drug Administration, Center for Drug Evaluation and Research, Silver Spring, Maryland^t; National Eczema Association, Novato, California^u; Departments of Dermatology and Pediatrics, University of California, San Diego School of Medicine, San Diego, California^v; Rady Children's Hospital-San Diego, San Diego, California^w; Department of Dermatology, Eastern Virginia Medical School, Norfolk, Virginia^x; Brigham and Women's Hospital, Boston, Massachusetts^y; Department of Dermatology, Harvard Medical School, Boston, Massachusetts^z; and Department of Dermatology, Medical University of South Carolina, Charleston, South Carolina.^{aa}

Funding sources: This work was supported by a generous grant from the Pediatric Dermatology Research Alliance (PeDRA), with additional funding from the National Eczema Association (NEA) and the National Psoriasis Foundation (NPF). Funding sources had no influence on the content of the guidelines beyond several authors being members of the funding sources and review and approval of the final document by the Pediatric Dermatology Research Alliance (PeDRA) Best Practices Task Force.

Disclaimer: This publication has been reviewed by FDA and determined not to be consistent with the Agency's views or policies. It reflects only the views and opinions of the authors.

Patient consent: Not applicable.

IRB approval status: Not applicable.

Key words: adolescent; child; consensus; guidelines; methotrexate; pediatric.

Correspondence to: Heather A. Brandling-Bennett, MD, Seattle Children's Hospital, 4800 Sand Point Way NE, M/S OC.9.835, Seattle, WA 98105

E-mail: heather.brandling-bennett@seattlechildrens.org

Conflicts of interest

Dr Brandling-Bennett has served as principal investigator for Lilly. Dr Arkin has received consulting fees from AbbVie, Regeneron, and Verrica. Dr Hebert has received research grants (paid to UTHealth McGovern Medical School) from LEO Pharma, Mayne Pharma, UCB, GSK, Ortho Dermatoligics, Amgen, Arcutis, AbbVie, and Pfizer; has received honoraria from Pfizer, Arcutis, UCB, Verrica, LEO Pharma, Novan, Incyte, Janssen, Mayne Pharma, and Castle Creek Bioscience; and has served on drug safety monitoring boards for Ortho Dermatologics, GSK, and Sanofi Regeneron. Dr Cordoro has served as a member of a scientific steering committee for Celgene. Dr Curran has served as an advisory board member for Novartis, receiving honoraria. Dr Dalrymple has served as an advisory board member for Novartis. Dr Flohr is chief investigator of the UK National Institute for Health Research-funded TREAT (ISRCTN15837754, EudraCT 2,015-002,013-29) and SOFTER (Clinicaltrials.gov: NCT03270566) trials as well as the UK-Irish Atopic Eczema Systemic Therapy Register (A-STAR; ISRCTN11210918) and a principal investigator in the European Union (EU) Horizon 2020-funded BIOMAP Consortium (<http://www.biomap-imi.eu/>). He also leads the EU Trans-Foods Consortium. His department has received funding from Sanofi-Genzyme and Pfizer for host-microbiome research. Dr Gordon has received personal fees from AbbVie, Almirall, Amgen, BMS, Dermavant, Dermira, Lilly, Janssen, Kyowa Hakko Kirin, LEO Pharma,

Novartis, Pfizer, Sun Pharma, and UCB and grants from BMS, Lilly, and UCB. Dr Irvine has received person fees from AbbVie, LEO Pharma, Lilly, Novartis, and Sanofi/Regeneron, and has received a grant from the UK National Institute for Health Research (research arm of the UK Department of Health) as co-principal investigator of a clinical trial comparing MTX with cyclosporine in children with severe atopic dermatitis. Dr Kirkorian has received honoraria from Verrica. Dr Lara-Corrales has served as an advisory board member for Ipsen, LEO Pharma, Lilly, Novartis, Pfizer, and Sanofi Genzyme; as a consultant for AbbVie, Avicanna, Janssen, Johnson & Johnson, L'Oréal, Pfizer, Pierre Fabre, Sanofi Genzyme, and Valeant; as a speaker for AbbVie, Amgen, Novartis, and Sanofi Genzyme; and has served as a site investigator for AbbVie, Clementia Pharmaceuticals, Janssen, Lilly, and, Mayne Pharma. Dr Jill Lindstrom has received consulting fees from AbbVie, AnaptysBio, Arena Pharmaceuticals, Aristeia, Incyte, and Priovant. Dr Amy S. Paller has served as an investigator for AbbVie, Applied Pharma Research, Dermavant, Eli Lilly, Incyte, Janssen, Krystal, Regeneron, and UCB; Consultant for Aegerion Pharma, Azitra, BioCryst, Boehringer-Ingelheim, Bristol Myers Squibb, Castle Creek, Eli Lilly, Janssen, Krystal, LEO Pharma, Novartis, Regeneron, Sanofi/Genzyme, Seanergy, TWI Biotechnology, and UCB, and on the data safety monitoring board for AbbVie, Abeona, Catawba, Galderma, and InMed. Author Begolka has served as an advisory board member for Incyte and Pfizer, and has received a grant from Pfizer. Dr Tom has served as a site investigator for clinical trials for Janssen, Regeneron, Incyte, Pfizer, Dermira, Eli Lilly, and AbbVie, and has been on data safety committees for LEO Pharma. Dr Voorhees has received research grants from AbbVie, Celgene, and Lilly; has served as an advisory board member for BMS, Boehringer Ingelheim, Celgene, Novartis, and UCB; and had received consulting fees from Amgen. Dr Vleugels has received consulting fees from AbbVie, Pfizer, and Priovant. Dr Lee has received consulting fees from AbbVie and Krystal Biotech; served as an investigator for AbbVie, Amryt, Arcutis, BMS, Castle Creek Biosciences, Celgene, Galderma, Incyte, Janssen, Kiniksa, Lilly, Mayne Pharma, Novartis, Pfizer, Regeneron, Sanofi Genzyme, Target Pharma, Trevi Therapeutics, and UCB; and has served as an advisory board member for Lilly, Novartis, Pfizer, and Regeneron. Dr Davies has received a fellow stipend from Pediatric Dermatology Research Alliance (PeDRA). Dr Siegfried has received consulting fees from AbbVie, Boehringer Ingelheim, Incyte, LEO Pharma, Novan, Novartis, Pierre Fabre, Pfizer, Regeneron, Sanofi Genzyme, UCB, and Verrica; has received honoraria from Regeneron, Sanofi Genzyme, and Verrica; has served on data safety monitoring boards for LEO Pharma, Novan, Pfizer, and UCB; participated in contracted research for AI Therapeutics; served as principal investigator for Janssen, and her institution has received fees related to clinical trials from Janssen, Lilly, Pierre Fabre, Regeneron, and Verrica and in support of a 2020-2021

pediatric dermatology fellowship from Pfizer. Dr Chiu, Dr Callen, Dr Castello-Soccio, Dr Co, Author Hanna, Dr Kim, and Dr Reyes have no conflicts of interest to declare.

REFERENCES

1. Siegfried EC, Arkin LM, Chiu YE, et al. Methotrexate for inflammatory skin disease in pediatric patients: consensus treatment guidelines. *Pediatr Dermatol*. 2023;40(5):789-808. <https://doi.org/10.1111/pde.15327>
2. Ebell MH, Siwek J, Weiss BD, et al. Strength of recommendation taxonomy (SORT): a patient-centered approach to grading evidence in the medical literature. *J Am Board Fam Pract*. 2004;17(1):59-67. <https://doi.org/10.3122/jabfm.17.1.59>

<https://doi.org/10.1016/j.jaad.2023.12.074>

Combined reflectance confocal microscopy and optical coherence tomography in the evaluation of cutaneous squamous cell carcinoma



To the Editor: Clinical differentiation between actinic keratosis (AK), squamous cell carcinoma in situ (SCCIS), and invasive SCC can often be challenging because early lesions have a similar appearance. Reflectance confocal microscopy (RCM) is a noninvasive imaging modality that can be used in vivo to produce high-resolution images but is limited to a depth of 250 μm in the skin.¹ In contrast, optical coherence tomography (OCT) can view deeper levels of skin (up to 2000 μm) but with a low resolution.² Therefore, combining the higher cellular resolution of RCM and the depth of OCT in one technology helps overcome these limitations.

We performed a prospective study, enrolling consecutive patients with AK, SCCIS, and invasive SCC, from September 2020 to December 2020. AK and SCCIS were grouped under one category for the feasibility of analysis. Patients enrolled had the following: (1) clinically suspicious lesions for SCC or SCCIS/AK—imaging was performed before the biopsy—or (2) biopsy-proven SCC or SCCIS that were referred for Mohs micrographic surgery (MMS)—imaging was performed before MMS to check for residual disease and margin mapping, and imaging operator was blinded to the biopsy results while imaging. Well-documented RCM and OCT diagnostic features of AK and SCC in the literature were used in this study (Figs 1 and 2).³⁻⁵ RCM-OCT findings were correlated with histopathology results including the presence and type of tumor, depth, and surgical margins.

There were 44 lesions imaged from 37 patients. Twenty-three lesions were referred for